Sarjy

VOICE · MEMORY · INTELLIGENCE

Product Requirements Document

A full-stack AI voice assistant built for the Sarj take-home assessment

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VERSION
2.0 — Ambitious

DATE
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Next.js 15 · Claude Haiku · Groq Whisper · ElevenLabs · Supabase · Upstash Redis · Twilio · PWA

Overview

Sarjy is a production-grade AI voice assistant with persistent memory, real-time tool execution, multilingual support, and telephony — designed to feel like months of work in a remarkably short build cycle.

> *"It should feel like talking to someone who actually knows you — and gets smarter every conversation."*

1 Problem Statement

> Existing voice assistants (Siri, Alexa) are stateless — they forget everything between sessions, don't support natural conversation memory, and can't be extended with custom tools or workflows. Sarjy solves all three.

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No Memory

Current assistants start fresh every session. Users repeat themselves constantly.

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High Latency

Most voice pipelines take 3–5s end-to-end. Sarjy targets sub-1s first audio.

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No Extensibility

Consumer assistants can't be given custom tools, workflows, or integrations.

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Platform Lock-in

Tied to specific devices. Sarjy is web-first and works via phone call too.

2 Goals & Success Metrics

Goal	Metric	Target
Low latency	Time to first audio byte after speech ends	< 1.2s
Memory accuracy	Correct recall of facts stated 3+ sessions ago	100%
Tool reliability	Successful tool execution rate	> 95%
Multilingual	Languages supported without config changes	90+ (Whisper)
Uptime	Vercel edge availability	99.9%
Cost efficiency	Cost per conversation	< $0.003

3 System Architecture

> All components run server-side on Vercel serverless functions. The client is a stateless Next.js app — no sensitive keys are exposed to the browser.

Core Voice Pipeline

🎤 Mic → VAD (silero_vad_v5) → WAV Blob → /api/stt (Groq Whisper) → Transcript

Transcript → /api/chat (Claude Haiku) → SSE Stream → /api/tts (ElevenLabs) → 🔊 Audio

Tool Calls → Weather API | Tavily Search | Google Calendar | Slack

Memory Architecture

Short-term — Upstash Redis

Last 20 messages per user. 24-hour TTL. Retrieved on every request to give Claude conversation context. Sub-5ms reads from Vercel edge.

Long-term — Supabase Postgres

Persistent facts: name, preferences, todos, reminders. Written by Claude via `save_memory` tool. Row-level security ensures per-user isolation.

4 Technology Stack

Layer	Technology	Why
Frontend	Next.js 15, React 19, Tailwind	App Router, RSC, streaming-native
STT	Groq Whisper large-v3-turbo	~300ms, 90+ languages, free tier

Layer	Technology	Why
LLM	Anthropic Claude Haiku 4.5	Fastest Claude, tool use, streaming
TTS	ElevenLabs eleven_turbo_v2_5	~300ms first chunk, most natural voice
VAD	@ricky0123/vad-web (Silero v5)	Browser-native, no server round-trip
Auth	NextAuth + Google OAuth	Reuse Google token for Calendar
Short memory	Upstash Redis	Serverless, edge-native, TTL support
Long memory	Supabase Postgres	RLS, structured queries, free tier
Telephony	Twilio Voice + WebSockets	Real phone number, bidirectional audio
Search	Tavily API	LLM-optimised results, generous free tier
Deploy	Vercel	Zero-config Next.js, edge runtime

5 Feature Requirements

Feature	Description	Priority	Status
Voice Input (VAD)	Auto speech detection — no push-to-talk. Silero VAD v5 runs in browser AudioWorklet.	P0	Done
Speech-to-Text	Groq Whisper large-v3-turbo. Auto language detection. ~300ms transcription.	P0	Done
LLM Chat	Claude Haiku with streaming SSE. Tool use with visible badges in transcript.	P0	Done
Text-to-Speech	ElevenLabs Turbo v2.5 streamed via Web Audio API with AnalyserNode amplitude tracking.	P0	Active
Persistent Memory	Claude auto-saves facts via tool. Two-tier: Redis (24h context) + Supabase (permanent).	P0	Done
Memory Panel	Slide-out drawer showing all saved facts grouped by category. Live updates. Delete individual or all.	P0	Done
Weather Tool	Real-time weather via OpenWeatherMap. Returns temp (F/C), condition, humidity, wind.	P0	Done
Web Search Tool	Tavily API for real-time information. LLM-optimised result summaries.	P0	Done
Interrupt Detection	VAD onSpeechStart stops TTS playback instantly. Natural barge-in behaviour.	P1	Next
Streaming TTS (sentence-level)	Split Claude output at sentence boundaries, pipe each to ElevenLabs immediately. First audio in ~800ms.	P1	Next
Language Switching	Whisper detects spoken language. Claude responds in same language. Flag shown in status bar.	P1	Next

Feature	Description	Priority	Status
Animated Lip-Sync Avatar	SVG face animated by AnalyserNode amplitude. Mouth opens/closes with speech in real time.	P1	Next
Google Calendar Integration	Read/create events via reused Google OAuth token. "What's on my calendar tomorrow?"	P1	Next
Personality Assessment Workflow	5-question MBTI-style flow guided by voice. Result saved to memory. Informs future responses.	P1	Next
Guardrails	System prompt topic blocklist + Llama Guard pre-check on Groq. Graceful deflections.	P1	Next
Telephony (Twilio)	Inbound phone number → bidirectional WebSocket → full voice pipeline over PSTN.	P2	Planned
Multimodal Input	Image upload button. Claude Vision describes image. "What's in this photo?"	P2	Planned
Slack Integration	Outbound webhook. "Sarjy, message #general that I'll be late."	P2	Planned
Mobile PWA	Installable on iOS/Android. Offline splash. Touch-optimised orb. Theme colour.	P2	Planned

6 Build Timeline

PHASE 0–7 · COMPLETE

Core Voice Pipeline

VAD → Groq STT → Claude Haiku (streaming SSE + tool use) → ElevenLabs TTS → two-tier memory → Google Auth → Weather + Search tools. Full UI with animated orb, transcript, memory panel, status bar.

PHASE A · IN PROGRESS

UX Upgrades — Interrupt, Streaming TTS, Language

Barge-in interrupt detection via VAD onSpeechStart. Sentence-level TTS streaming cuts latency to ~800ms. Language auto-detection from Whisper, Claude mirrors language, flag shown in status bar.

PHASE B

Animated Lip-Sync Avatar

SVG face driven by Web Audio AnalyserNode. Mouth morphs between open/closed states proportionally to audio amplitude. Already have AnalyserNode plumbed in from TTS player.

PHASE C

Integrations — Calendar, Slack, Flights

Google Calendar read/write via reused OAuth access token. Slack outbound webhook. Flight search via Tavily with structured query formatting. All exposed as Claude tool calls.

PHASE D

Multistep Workflows

Personality assessment: 5-question voice MBTI, result saved to Supabase memory. Symptom intake: structured chief complaint → duration → severity → summary. Both driven by conversation state, no extra code.

PHASE E

Guardrails + Multimodal

Llama Guard pre-check on Groq before Claude. Topic blocklist in system prompt. Image upload button with Claude Vision for photo understanding.

PHASE F · STRETCH

Telephony — Twilio Phone Number

Twilio inbound voice call → TwiML Media Streams → WebSocket server → bidirectional audio → full Sarjy pipeline over PSTN. Demo: call a real phone number and talk to Sarjy.

PHASE G · FINAL

Deploy + Present

Vercel production deploy. Google OAuth redirect URIs updated. Loom recording. 5-minute demo script practiced. Cost analysis slide prepared.

7 Cost Model

> Per-conversation cost breakdown based on a typical exchange: ~500 input tokens + ~200 output tokens + 10s audio input + 15s audio output. All prices as of March 2026.

Groq Whisper (STT — 10s audio)	`$0.000 (free tier)`
Claude Haiku 4.5 (500 in + 200 out tokens)	`$0.0005`
ElevenLabs Turbo v2.5 (~80 chars output)	`$0.0002`
Upstash Redis (per request)	`$0.000002`
Supabase (per read/write)	`$0.000`
Total per conversation	**`~$0.0007`**

Scale	Daily conversations	Monthly cost	With 10% margin
100 users	500 (5 avg/user/day)	~$10.50	~$12
1,000 users	5,000	~$105	~$120
10,000 users	50,000	~$1,050	~$1,200

At 10,000 users the biggest lever is ElevenLabs — switching to a self-hosted TTS model (e.g. Kokoro) could cut TTS cost to near zero. STT could move to Whisper.cpp on a dedicated GPU instance at ~$0.40/hr, covering ~80,000 minutes/hr of audio.

8 Latency Breakdown

Step	Current	Optimised (Phase A)
VAD end-of-speech detection	~200ms	~200ms (hardware-limited)
STT (Groq Whisper)	~300ms	~300ms
LLM first token (Claude Haiku)	~400ms	~400ms
TTS first chunk (ElevenLabs)	After full response (~1.5s)	After first sentence (~300ms)
Total to first audio	**~2.7s**	**~1.2s**

9 Risks & Mitigations

Risk	Likelihood	Mitigation
ElevenLabs API outage	Low	Fallback to browser Web Speech API `speechSynthesis`
Anthropic rate limits	Medium	Groq llama-3.3-70b as hot fallback (same tool schema)
VAD false triggers in noise	Medium	Discard audio clips < 0.5s before sending to STT
Google OAuth token expiry for Calendar	High	Store refresh token in Supabase, auto-refresh on 401
Supabase RLS misconfiguration	Low	All writes go through service-role server route, tested in migration

10 5-Minute Demo Script

> Audience: mostly technical, some non-technical. Goal: show it working end-to-end, then go deep on one impressive technical detail per minute.

MINUTE 0:00 – 0:45

Sign in + Stack Overview

Open the app cold. Sign in with Google. While it loads, explain the stack in one sentence: "Next.js frontend, Groq for speech-to-text, Claude Haiku for intelligence, ElevenLabs for voice, with two-tier memory in Redis and Postgres." Show the sign-in card design.

MINUTE 0:45 – 1:30

Memory — the killer feature

Say: "Hi Sarjy, my name is [name] and I'm a software engineer who loves coffee." Open the memory panel — watch the facts appear live. Reload the page. Ask: "What do you remember about me?" — Sarjy recalls everything. This proves persistence.

MINUTE 1:30 – 2:30

Tool Calls — Weather + Calendar

Ask: "What's the weather in London right now?" — show the tool badge appear in the transcript, then Sarjy speaks the answer. Then: "What's on my calendar tomorrow?" — show Google Calendar tool call. Open Network tab to show the SSE stream in real time.

MINUTE 2:30 – 3:15

Multilingual + Interrupt

Speak in another language — Sarjy auto-detects and responds in kind. Then start speaking while Sarjy is mid-sentence — audio cuts off instantly (barge-in). Explain: "Whisper detects language, Claude mirrors it — no configuration changes needed."

MINUTE 3:15 – 4:15

Personality Assessment Workflow

Say: "Sarjy, let's do a personality assessment." Watch Claude guide through 5 voice questions. Result appears in memory panel. Explain: "Multi-step workflows are pure conversation state — no extra code, just Claude knowing when to proceed."

MINUTE 4:15 – 5:00

Telephony + Q&A Setup

Call the Twilio number from a real phone. Sarjy answers and has the same conversation over PSTN. Close with cost model: "$0.0007 per conversation — at 10,000 users, ~$1,200/month. Main lever to scale down: self-hosted TTS." Open for questions.

11 Questions to Prepare

- **Why Claude Haiku over GPT-4o-mini?** — Tool streaming format is cleaner, SSE native, and we were already on Anthropic infrastructure.
- **How do you handle hallucinations?** — Guardrails via Llama Guard pre-check + system prompt topic list. Memory facts are structured (key/value), not free text, so Claude can't hallucinate what was saved.
- **What happens if the user speaks in background noise?** — Silero VAD handles it at the model level. We also discard <0.5s clips before STT. In telephony, Twilio's noise suppression handles PSTN noise.
- **How would you scale this to 1M users?** — Redis → Redis Cluster, Supabase → connection pooling via PgBouncer, TTS → self-hosted Kokoro model, STT → Whisper.cpp fleet, LLM → batch routing with priority queues.
- **Why not use a unified voice API like Vapi/Retell?** — Control and cost. Sarjy's pipeline is 4× cheaper per conversation and fully customisable — you can swap any component without vendor lock-in.